Automated Tunneling Nanotube Detection Report

1 Literature Review

Table 1: Paper Review Summary

Title	Year	Data	Code/Model	Method
Automated detection of tunneling nanotubes in 3D images	2006	No	No	Watershed and edge detection
TNTdetect.AI: A Deep Learning Model for Automated Detection and Counting of Tunneling Nanotubes in Microscopy Images	2022	No	No	U-Net
Detection and Quantification of Tunneling Nanotubes in Multidimensional Bioimage Data	2025	Yes	Napari Plugin	Mathematical Morphology
Segmentation of Tunneling Nanotubes Using U-Net-Like Neural Networks	2025	Yes	TNTSeg Python Library	U-Net

The two 2025 master's theses are the most relevant, and both were supervised by Martin Maška (associate professor, Masaryk University, Brno), who has published with Jean-Yves Tinevez. The first ships a napari plugin based on mathematical morphology; it is GPU-only (CuPy/CUDA), so on macOS it requires a CPU fallback and runs slowly. The second ships the `TNTSeg` library with annotated images and trained U-Net models. Its main limitations are that the images are low resolution, the models segment every z-stack without applying TNT-specific criteria, and they are trained on fixed 7-plane stacks, so they do not generalise to volumes with a different number of z-planes.

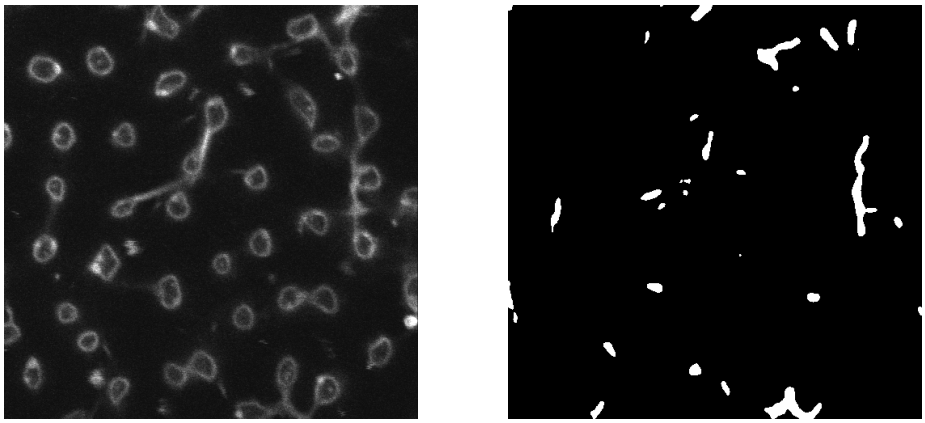


Figure 1: A TNTSeg example image (left) and its binary mask result (right).

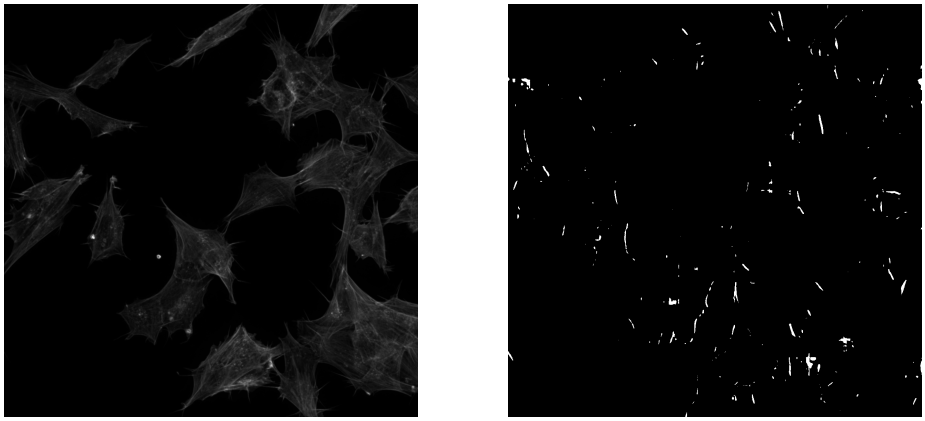


Figure 2: One of our ND2 images (left) and its binary mask from TNTSeg (right).

2 Data from Membrane Traffic & Pathogenesis Lab

Two 3D confocal datasets were provided (Table 2). Ground truth was annotated in Icy: the czi set has per-TNT annotations for all 10 images, so detections can be matched to individual TNTs; the nd2 set gives only per-image TNT counts, for 6 of the 7 images.

Table 2: Provided datasets.

Set (folder)	Format	Files	Channels	Size (px)	Z / voxel xy,z (µm)	Ground truth
Zeiss (`Rapamycin DsRed...`)	`.czi`	10	DAPI, AF488, AF555, AF647	2281^2	10–35 / 0.071, 0.35	per-TNT (503 cells, 52 TNTs)
Nikon (`Anna images-TNTs`)	`.nd2`	7	DAPI, TRITC, Cy5	1200^2	17–48 / 0.183, 0.40	counts, 6/7 (280 cells, 40 TNTs)

3 Experiments

My unsupervised pipeline runs in four steps (membrane channel: AF647 for czi, TRITC for nd2):

1. **Cells from DAPI.** Cellpose on the DAPI max-projection gives one label per nucleus, so the label count is the cell count. Each nucleus is then grown into a cell territory over the membrane channel. (DAPI is used because on the membrane channel touching cells merge and cells joined by a TNT are exactly the ones most likely to touch.)

2. **Candidate tubes.** On the membrane channel, a 1.2 µm white top-hat keeps thin bright structures, and a probabilistic Hough transform extracts straight-line candidates.

3. **Two-cell, not pre-existing.** A line is kept only if it runs *between exactly two* territories, each end near a different cell, and the pair is *not* already connected in the substrate planes below the cell layer

4. **Ridge test.** A true tube is brighter than the thin bands beside it; this bilateral ridge contrast is scored in σ, and only pairs above threshold are reported.

Cell counts (Cellpose) and detected TNTs were then compared against the provided ground truth (Tables 3 and 4).

Table 3: Zeiss (czi) results. CP = Cellpose, GT = ground truth

Image	Cells CP	Cells GT	TNT det.	TNT GT
1	47	46	15	3
2	61	59	12	7
3	46	41	17	7
4	56	53	17	3
5	62	62	39	8
6	58	57	20	2
7	55	60	16	9
8	50	49	10	5
9	37	36	14	2
10	41	40	6	6
Total	**513**	**503**	**166**	**52**

Table 4: Nikon (nd2) results. Counts only, no annotated coordinates; wt002 is unannotated. Total is over the 6 annotated images.

Image	Cells CP	Cells GT	TNT det.	TNT GT
wt001	57	51	11	8
wt002	49	–	8	–
wt006	75	61	17	7
wt007	56	47	12	7
wt008	48	47	6	10
wt009	50	46	4	5
wt010	40	28	4	3
Total (6)	**326**	**280**	**54**	**40**

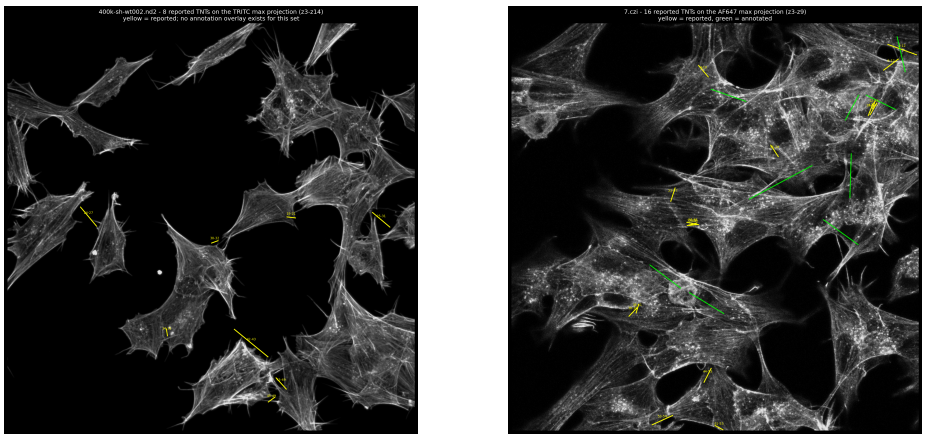


Figure 3: Hough candidates (yellow). Left: Nikon example (no ground truth). Right: Zeiss example with annotated TNTs in green.

Although the method produces candidates, they often do not fall in the expected locations. Some results are promising, but accuracy is generally limited

Several unsupervised methods were tried for preprocessing (intensity/histogram projection and background subtraction) and detection (Canny edges, Hough lines). Hough lines works best so far and is promising, but a supervised approach is likely to be more reliable.

4 Future Directions

Annotation is the priority. A **bounding box** marks each TNT with a rectangle; a **mask** labels the exact TNT pixels. Suitable 3D annotation tools exist, and Jean-Yves Tinevez could advise on the most microscopy-friendly option.

With annotations, we could train our own deep-learning model to segment or detect TNTs quickly and accurately across images from different sources.